

Autodesk® Scaleform®

AMP

Scaleform Analyzer for Memory and Performance AMP

Alex Mantzaris
3.03
2013 4 10

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Autodesk Scaleform:

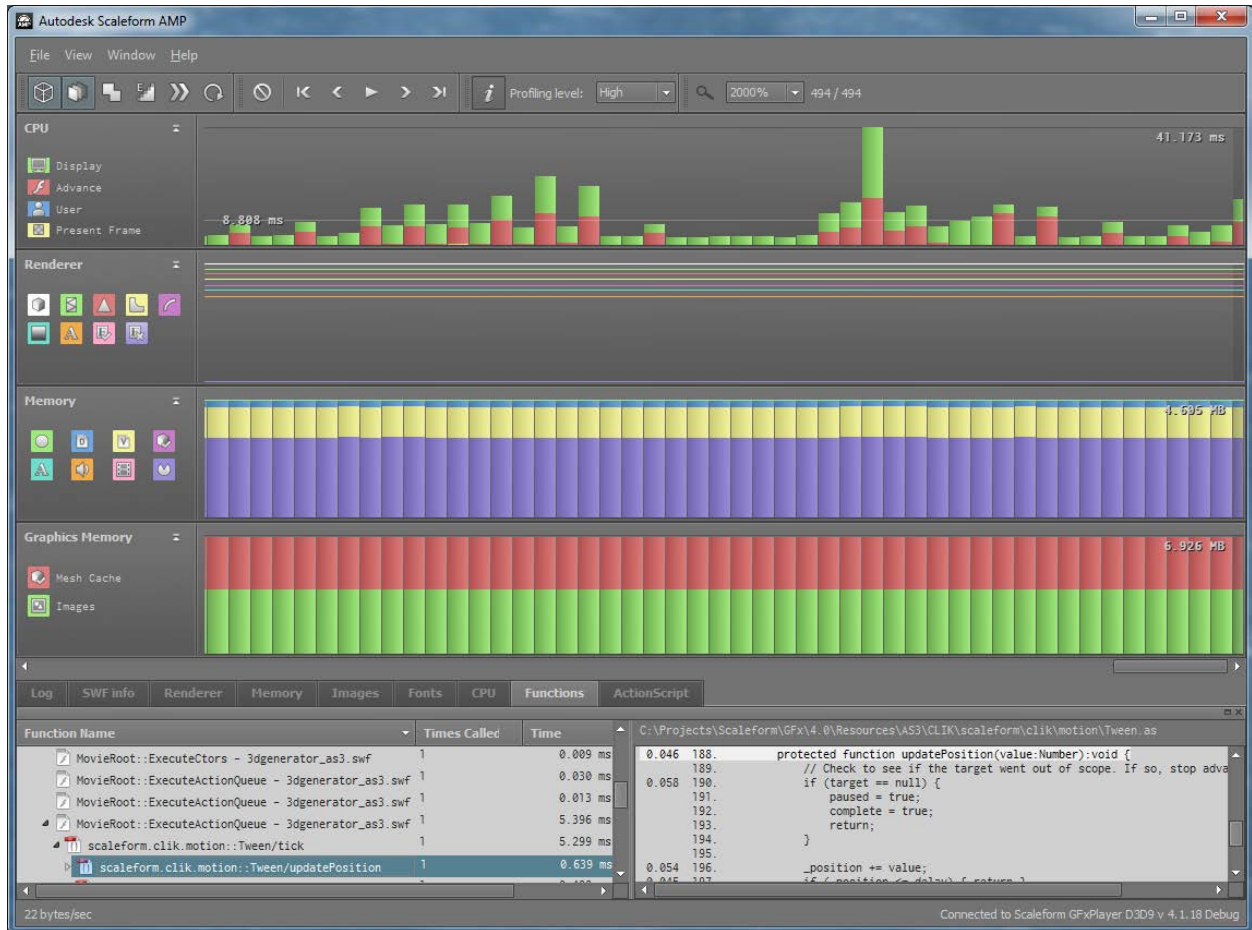
	AMP
	Autodesk Scaleform Corporation
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	Greenbelt, MD 20770, USA
	www.scaleform.com
Email	info@scaleform.com
	+1 (301) 446-3200
Fax	+1 (301) 446-3199

		1
1.1	AMP	2
1.2	AMP	3
		12
2.1		12
2.2		13
2.3		14
2.4		14
2.5		19
2.6		20
		22
		23
		24
5.1	Debug	24
5.2		24
5.3		25
5.4		25
		26
		27

AMP™ Scaleform

CPU
Scaleform®

ActionScript



Scaleform

AMP

AMP

AMP

AMP Scaleform 3.2

1.1 AMP サーバー

A word cloud of programming languages and technologies. The words are scattered across the page in various sizes and orientations. The most prominent words are 'ActionScript', 'Flash', 'AMP', 'Scaleform', 'C++', 'Movie', 'SWF', 'Movies', 'Flash', 'SWD', 'AS', and 'ActionScript3'.

1.2 AMP クライアント

AMP

AMP

Scaleform

Scaleform CLIK

AMP

Flash

- Scaleform (Display) Direct Access API (User) Scaleform CPU

- [Scaleform Best Practices Guide](#) 

Scaleform

- Scaleform

- AMP Scaleform GPU

- Log Scaleform

- Flash

- o *Text Fields optimized for readability:* Anti-alias for readability

Anti-alias for animation

- 1
0 *Ratio of draw primitives to number of meshes*

1.0 1 1

- o *Number of draw primitives*

- o *Number of meshes*
- o *Number of triangles:*
- o *Mesh cache thrashing:*

- o *Number of strokes*
Flash
- o *Number of gradient fills*

- o *Number of masks*
- o *Number of filters*
- o *Number of DirectAccess API calls* API
- o *Number of rasterized glyphs*
- o *Number of font textures* 1
- o *Number of font texture updates*

- o *Font cache thrashing*

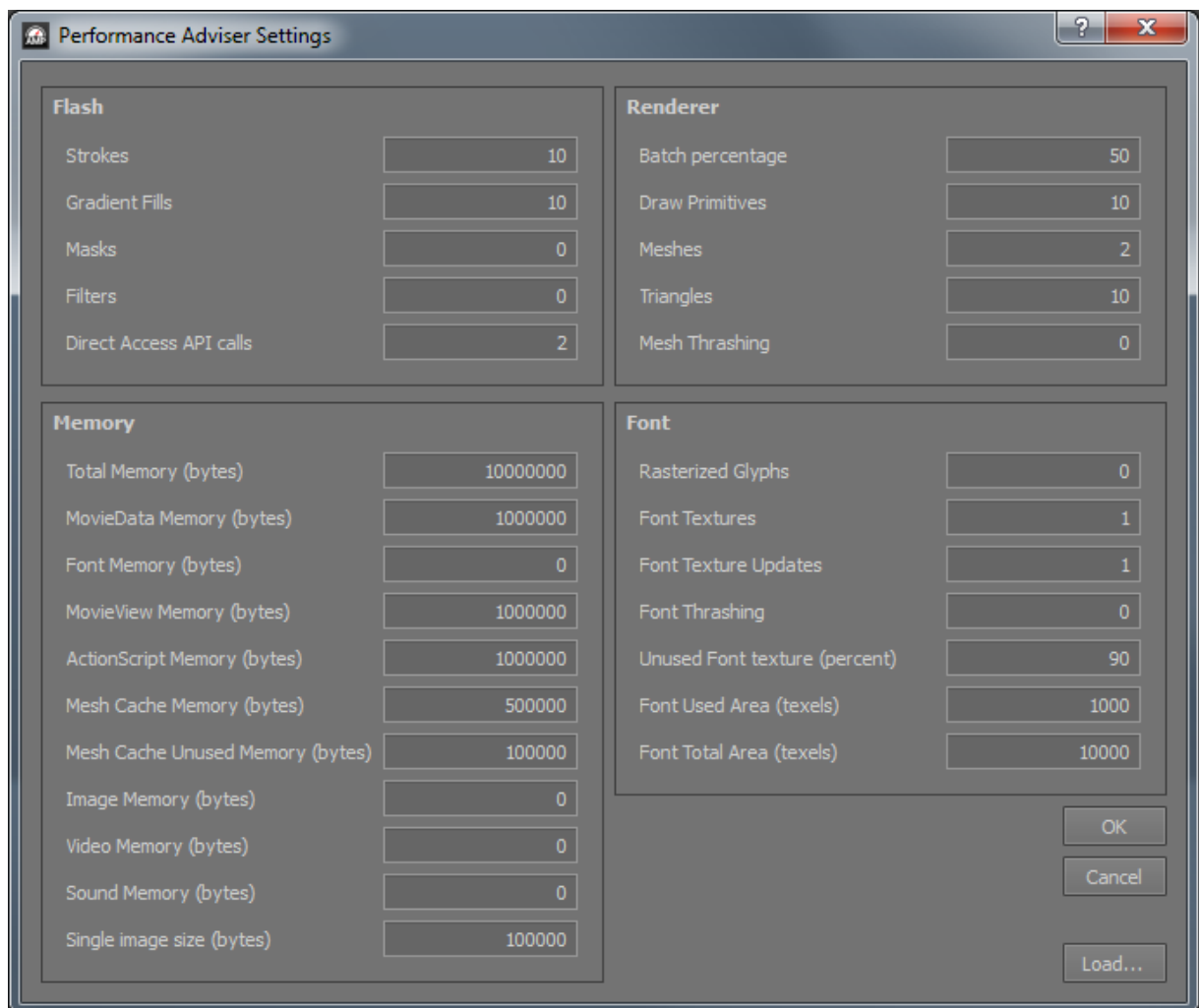
- o *Unused font texture percentage*

- o *Used font cache area*
- o *Total font cache area*

- o *Total Memory* Scaleform
- o *MovieData Memory*
- o *Font Memory (フォントメモリー) :*
- o *MovieView Memory* Gfx::Movie
MovieView
- o *ActionScript Memory* ActionScript
ActionScript

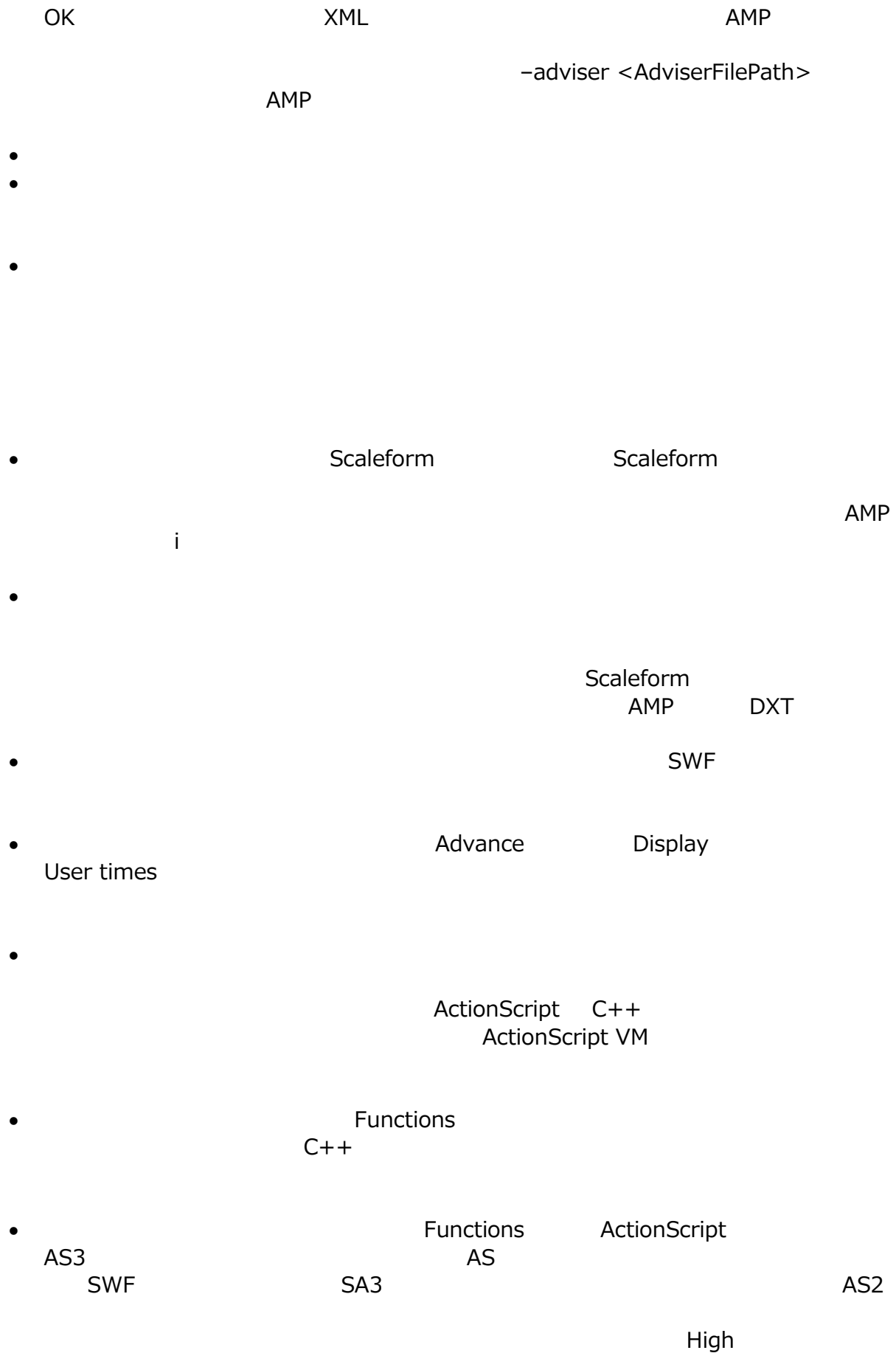
- o *Mesh Cache Memory*
- o *Unused Mesh Cache Memory*
- o *Image Memory*
- o *Video Memory* GPU
- o *Sound Memory*
- o *Number of images exceeding a certain size* AMP

adviser.xml XML AMP
Settings Performance Advisor



Section	Setting	Value
Flash	Strokes	10
	Gradient Fills	10
	Masks	0
	Filters	0
	Direct Access API calls	2
Memory	Total Memory (bytes)	10000000
	MovieData Memory (bytes)	1000000
	Font Memory (bytes)	0
	MovieView Memory (bytes)	1000000
	ActionScript Memory (bytes)	1000000
	Mesh Cache Memory (bytes)	500000
	Mesh Cache Unused Memory (bytes)	100000
	Image Memory (bytes)	0
	Video Memory (bytes)	0
	Sound Memory (bytes)	0
	Single image size (bytes)	100000
Renderer	Batch percentage	50
	Draw Primitives	10
	Meshes	2
	Triangles	10
	Mesh Thrashing	0
Font	Rasterized Glyphs	0
	Font Textures	1
	Font Texture Updates	1
	Font Thrashing	0
	Unused Font texture (percent)	90
	Font Used Area (texels)	1000
	Font Total Area (texels)	10000

Buttons: OK, Cancel, Load...



AS2

Scaleform

2

Flash

AMP

AMP

SWF

AS3

SWF

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0

0

0

AMP AS2 SWD
Add Remove

AS3

AS

0

0

Load Profile Frames

0

AMP

"AmpMemReport.txt"

0

ョンを終了します。

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AMP

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Memory

Memory

•

Images

Fonts

CPU

Functions

ActionScript

Log

SWF Info

Renderer

Memory

Restore UI

AMP

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Scaleform.com

2

User Guide

About

AMP

• A

UI

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AMP

Memory

AMP Images

10

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Low

Medium

High

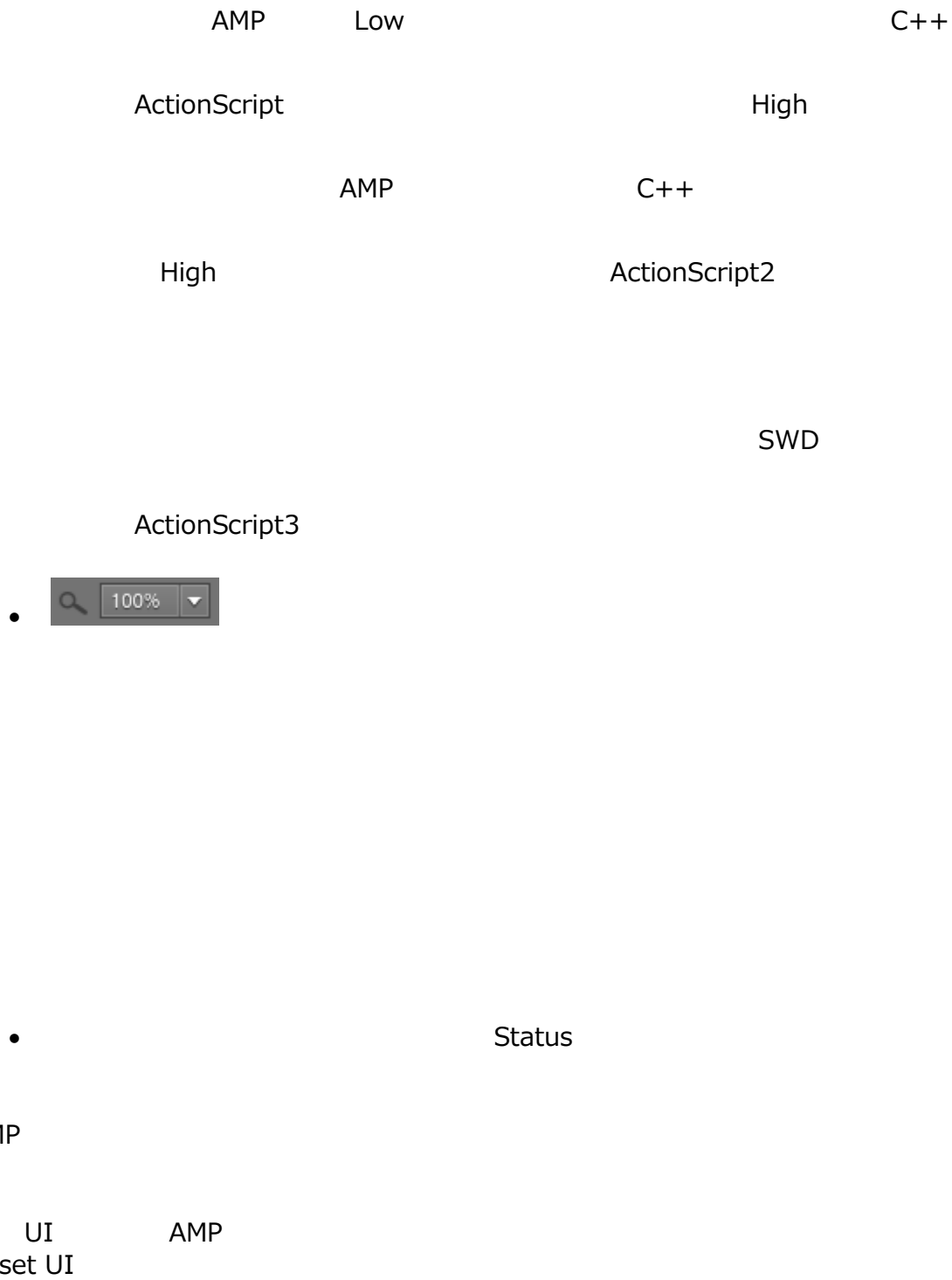
Low

AMP

C++

Medium

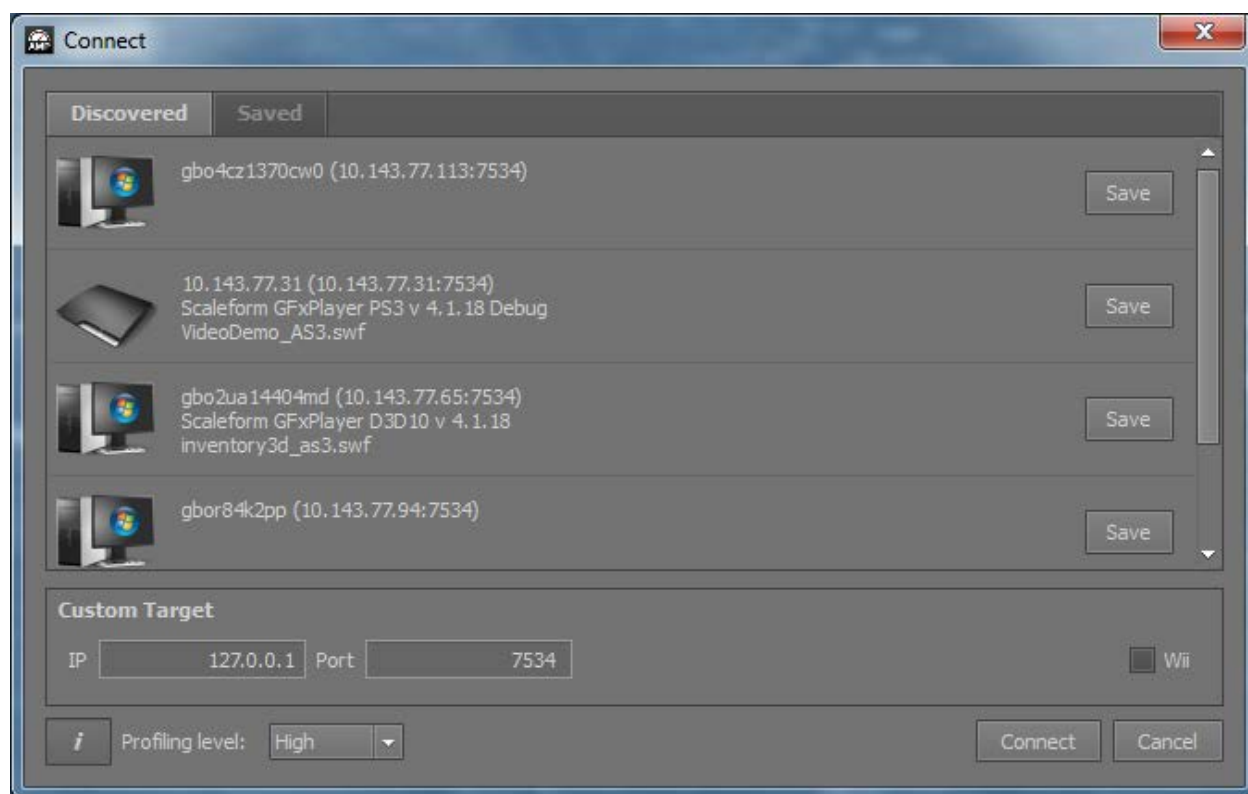
10



AMP
Bin/AMP Scaleform
GfxAmpClient.exe

2.1 コネクションの確立

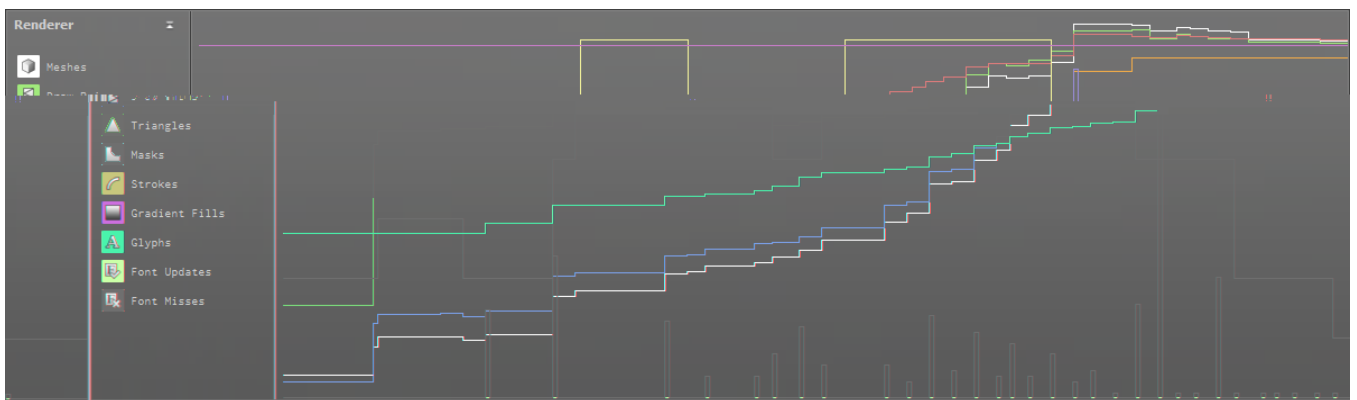
IP AMP
AMP



File Connection
AMP
Xbox360 PS3
IP Flash
AMP
Scaleform
Connect
AMP

2.3 レンダリング指標の解析

Best Practices Guide



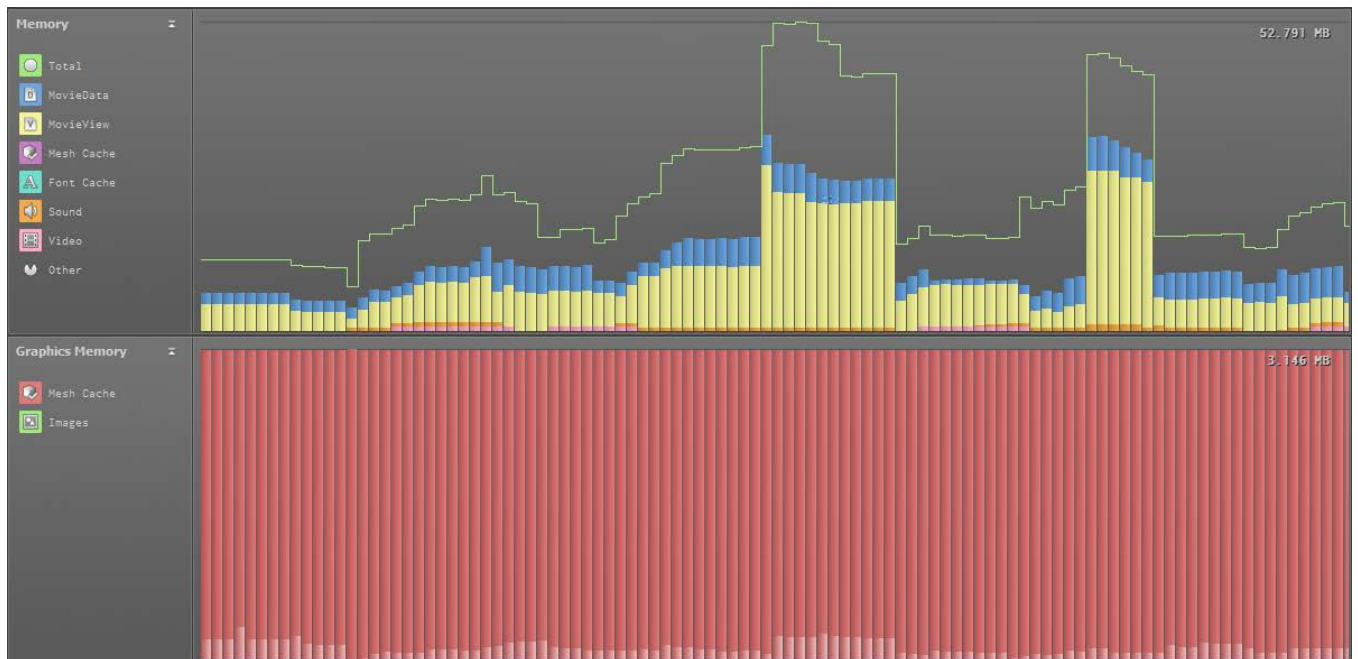
Log	SWF info	Renderer	Memory
Meshe		182	
Draw Primitives:		183	
Triangles:		5124	
Masks:		6	
Filters:		0	
Strokes:		21	
Gradient Fills:		0	
Rasterized Glyphs:		412	
Font Cache Updates:		0	
Font Textures:		1	
Font Cache Thrashing:		0	
Font Cache Fill:		28%	
Font Cache:		1048576 pixels	
Font Cache Used:		298294 pixels	
Font Failures:		0	
Font Cache Misses:		5	
Mesh Thrashing:		0	

2.4 メモリ消費の解析

3

Scaleform

4



- Scaleform
- SWF GFX
- GfX::Movie ActionScript
-
-
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- Scaleform

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Memory System Overview

“Used Space” GfX

Debug Data GfX
Memory

AMP

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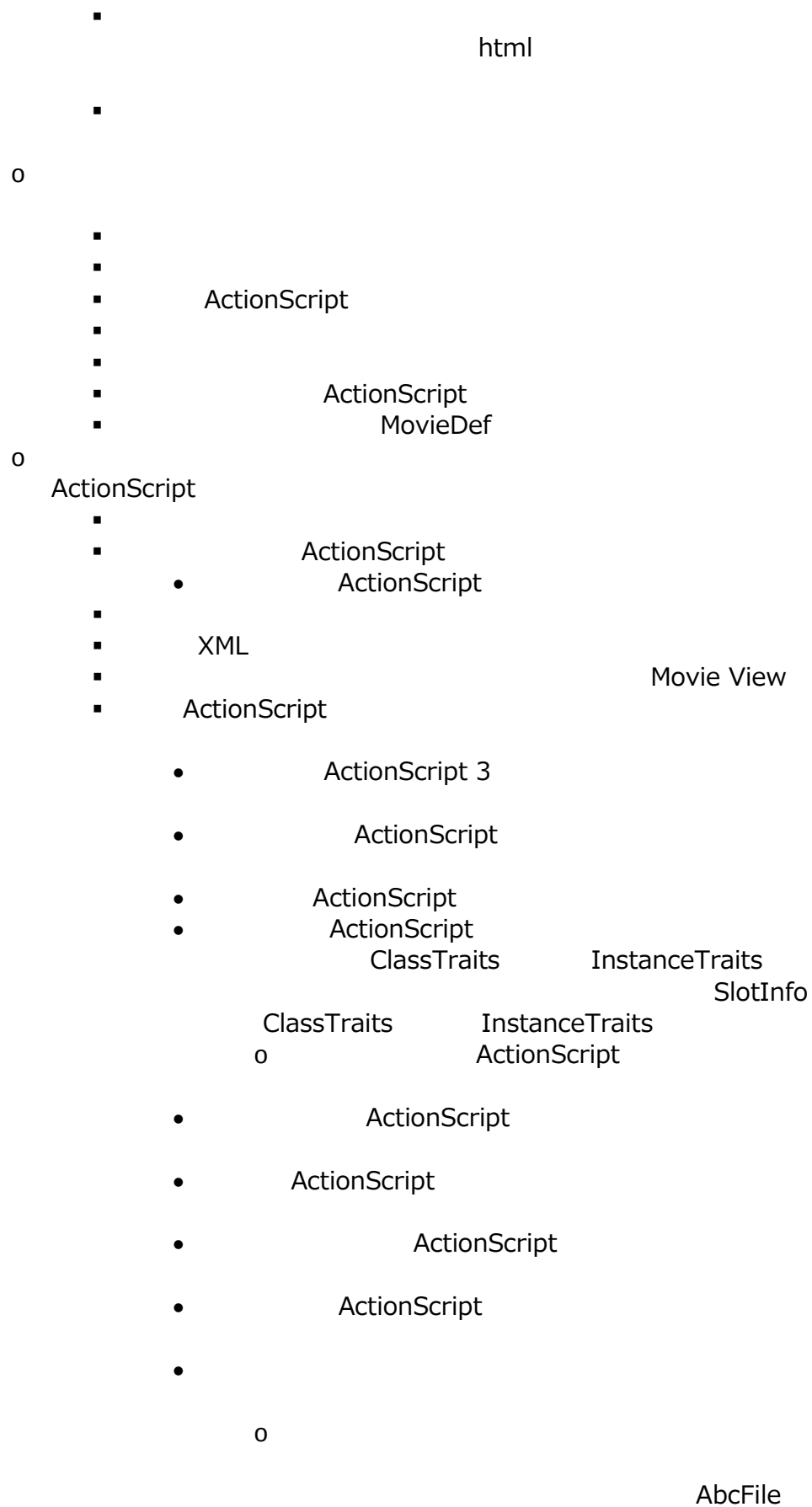
▪

▪

▪

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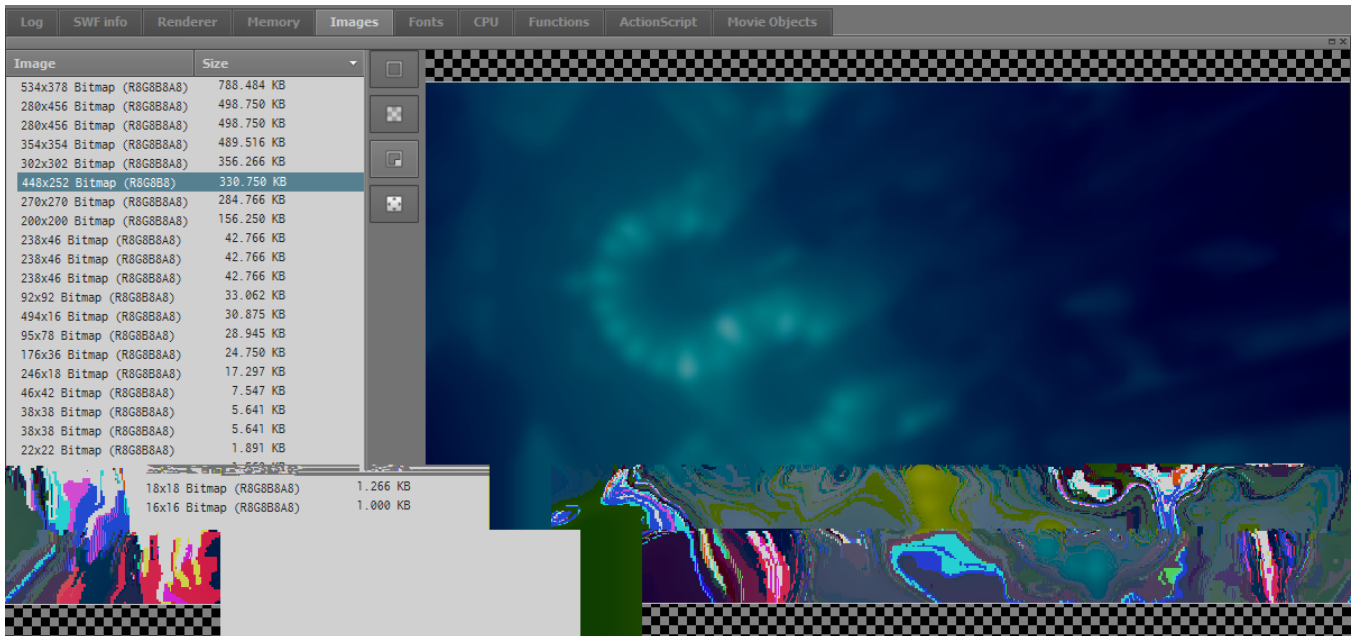
▪



- AbcFiles
-
- o Input Method Editor (IME)

Log	SWF info	Renderer	Memory	Images
Memory		Size		
▲	Total Footprint	4849.402 KB		
▲	Used Space	4633.715 KB		
▷	Video Heaps	0.000 KB		
▷	Other Heaps	325.621 KB		
▲	Movie View Heaps	1226.094 KB		
▲	MovieView "3dgenerator_as3.swf"	1226.094 KB		
	String	0.828 KB		
▷	Renderer	12.672 KB		
▷	MovieView	1176.781 KB		
	General	35.812 KB		
▷	Movie Data Heaps	261.066 KB		
▲	Global Heap	2820.934 KB		
	Video	264.176 KB		
	String	20.633 KB		
	Sound	0.141 KB		
▷	Renderer	2310.914 KB		
▷	MovieView	8.039 KB		
▷	MovieDef	116.676 KB		
	Image	0.141 KB		
	IME	9.914 KB		
	General	90.301 KB		
▷	Debug Memory	8.000 KB		
▷	Unused Space	38.820 KB		
	Debug Data	176.867 KB		

Scaleform



Scaleform

Scaleform

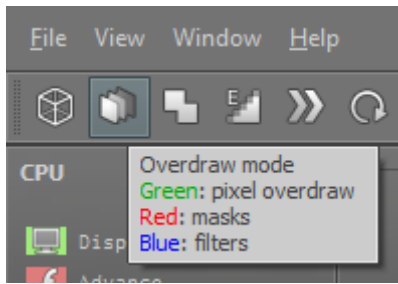
Fonts



SWF

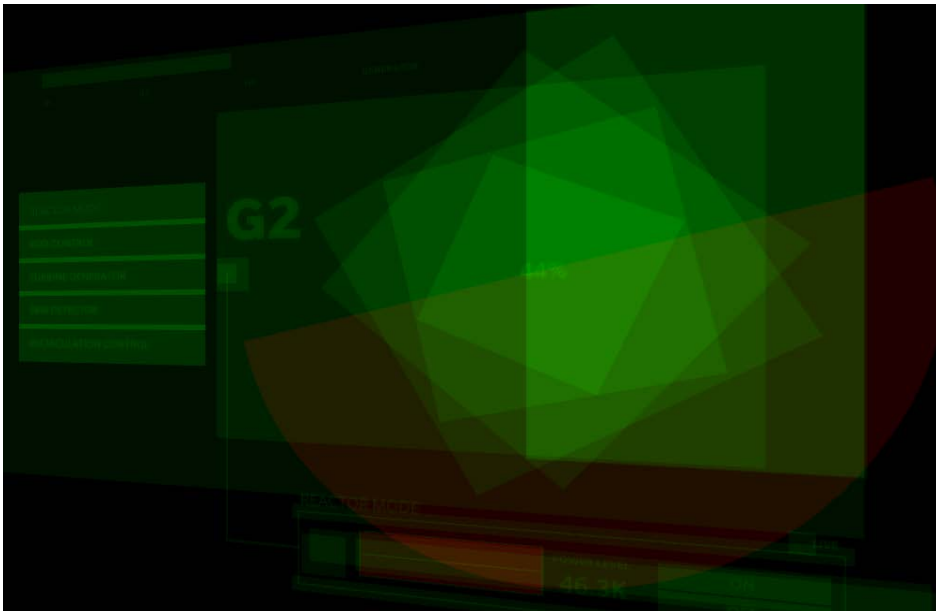
SWF
Heaps
AMP
MovieData
Memory Summary
CharDefs

2.5 オーバードロー プロファイリングの使用



16

8



2.6 バッチ プロファイリングの使用

AMP

1. AMP GFx::System Scaleform GFx::System::Init
 AMP Scaleform::System
 GFx::AMP::Server::Init() GFx::AMP::Server::Uninit()

2. AMP Windows Xbox360 PS3 Linux Mac Wii, Android
 iPhone/iPad, 3DS, PS Vita, WiiU AMP AMP IP
 IP

 - Xbox360 IP IP
 IP Xbox360 Neighborhood

 - PS3 2 IP
 IP IP

 - Android
 AndroidManifest.xml

 <uses-permission android:name="android.permission.INTERNET" />

 - WiiU WiFi PC
 PC 6003
 TCP/IP

 - Wii NDEV USB COM Wii
 NDEV hio.dll Wii AMP IP
 wii AMP
 RVL_SDK/X86/bin PATH

3. Unreal Engine 3
 Flash AS2 SWD AMP
 AMP SWF AMP
 SWD SWD
 AMP

1. AMP Scaleform
AMP GFxConfig.h GFX_AMP_SERVER
Scaleform Scaleform
AMP
Ptr<GFx::AMP::ServerState> serverState = *SF_NEW GFx::AMP::ServerState();
serverState->StateFlags &= GFx::AMP::Amp_Disabled;
AmpServer::GetInstance().UpdateState(serverState);

2. AMP

libnetctl Linux Windows Ws2_32.lib Xbox360 Xnet.lib PS3

3. AMP AMP

AMP

AMP
AMP::Server::GetInstance().SetInitSocketLib(false)
AMP

4. AMP Scaleform

AMP::Server::GetInstance().SetHeapLimit()

5. AMP 7534

AMP::Server::GetInstance().SetListeningPort()

6. AMP

AMP::Server::GetInstance().SetConnectionWaitTime

Scaleform

AMP

AMP

Scaleform Player

5.1 Debug ファイルの生成

AMP ActionScript 2 Flash

SWD

Flash CS3

CS4

Ctrl+Shift+Enter

SWD

SWD

SWF

AMP

ActionScript 3

SWD

SWF

AS

Flash Studio

SWF

5.2 アプリケーションのコントロール

AMP

AMP

app-control

- AMP::AppControlInterface app-control
 AMP::MessageAppControl AMP
 HandleAppRequest
- AMP::Server::GetInstance().SetAppControlCallback
- AMP::Server::GetInstance().SetAppControlCaps
 AMP true
 AMP::MessageAppControl

```
AMP::MessageAppControl caps;  
caps.SetCurveToleranceDown(true);  
caps.SetCurveToleranceUp(true);  
caps.SetNextFont(true);  
caps.SetRestartMovie(true);  
caps.SetToggleAaMode(true);  
caps.SetToggleAmpRecording(true);  
caps.SetToggleFastForward(true);  
caps.SetToggleInstructionProfile(true);  
caps.SetToggleOverdraw(true);  
caps.SetToggleBatch(true);
```

```
caps.SetToggleStrokeType(true);
caps.SetTogglePause(true);
caps.SetToggleWireframe(true);
AMP::Server::GetInstance().SetAppControlCaps(&caps);
```

5.3 コネクション・ステータス

AMP

```
AMP::Server::GetInstance().SetConnectedApp
```

AMP

- ServerStateType
 - AMP::Server::GetInstance().SetState
 - o Amp_RenderOverdraw
 - o Amp_App_Wireframe
 - o Amp_App_Paused
 - o Amp_App_FastForward
- AMP::Server::GetInstance().SetAaMode
- AMP::Server::GetInstance().SetStrokeType
- AMP::Server::GetInstance().SetCurrentLocale
- AMP::Server::GetInstance().SetCurveTolerance
- AMP::ServerState
 - AMP::Server::GetInstance().UpdateState

5.4 マーカー

C++

ActionScript

CPU

ActionScript

```
Amp.addMarker(1)
```

```
Amp.addMarker() .as
```

Resources/AS2/CLIK

Amp

C++

```
GfX::MovieImpl:: AdvanceStats->AddMarker(1)
```

AMP

AMP

GFX_AMP_SERVER

GF

AMP

Platform::RenderThread::drawFrame

AMP

AmpServer::GetInstance().AdvanceFrame

AMP

AMP

AMP

i

AMP

Platform::RenderThread::drawFrame

AMP

AmpServer::GetInstance().AdvanceFrame

AMP

1

Scaleform

AMP

AMP

i

AMP

profile level

1MB

Gfx::AMP::Server::SetHeapLimit

AMP

1MB

AmpServer::SetHeapLimit

AMP

GF

AMP

- [AMP Use Cases](#)
- [Scaleform Reference](#)
- [AMP Forum](#)